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First-generation Drones

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2.1 Introduction

The first generation of drones represents the foundational stage in the evolution of unmanned aerial systems (UAS), marking the earliest efforts to develop aircraft that could be controlled remotely without an onboard pilot. Emerging during the early to mid-20th century, these pioneering systems were primarily developed for military training, target simulation, and reconnaissance, long before drones became synonymous with modern intelligence and commercial applications. Their development was motivated by the need to reduce pilot risk in dangerous missions and to explore the potential of remotely piloted flight for defense and research purposes.

Technologically, first-generation drones were manually operated through rudimentary radio control systems, which transmitted limited signals from a ground station to the aircraft. These systems relied on basic mechanical servomechanisms and analog controls, which provided minimal precision and a short operational range. Lacking electronic stabilization, navigation sensors, or onboard processors, these early UAVs required continuous human control. The aircraft's performance and stability were heavily dependent on the operator's skill and line-of-sight visibility, as there were no telemetry systems to provide feedback during flight.